

Building a Khmer Traffic Sign Dataset for Automated Detection and Classification

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Abstract

Traffic Sign Detection and Classification have become a core feature of computer vision, contributing for intelligent transportation systems, traffic monitoring and driver assistance technologies. However, the availability of public datasets illustrating Khmer traffic signs remains limited, making it challenging to develop and evaluate reliable detection and classification models for Cambodia. To overcome the lack of the publicly available Khmer datasets, this research introduces a custom Khmer traffic sign datasets collected and annotated the capture image from various locations in Cambodia. The datasets contain an image captured under multiple real-world conditions including different lighting, different angles, distance, background and covers 42 classes commonly found in Cambodia. To evaluate the dataset, YOLOv8 and YOLOv10 models were trained and compared using multiple detection performance metrics. The results showed that YOLOv8 achieved slightly better overall performance across the majority of evaluation metrics, while both models demonstrated strong detection performance on the Khmer traffic sign dataset. The selected YOLOv8 model was further integrated into a bilingual web-based application supporting both Khmer and English languages for automated traffic sign detection and classification. This system demonstrates the practical application of deep learning-based Khmer traffic sign recognition and provides a foundation for future research and intelligent transportation applications in Cambodia.

Keywords: *Khmer Traffic Sign, Khmer Traffic Sign Detection and Classification (KTSDC), Computer Vision, Deep Learning*

1 Introduction

Traffic Signs play a significance role in modern transportation systems by providing an impor-

tance information, guiding driver behavior and enhancing road safety [1][2]. With the rapid development of computer vision and deep learning techniques, automated traffic sign detection and classification have become main research areas for intelligent transportation systems and driver assistance technologies.

Developing accurate and reliable traffic sign detection and classification models require a large-scale dataset. Existing datasets, including German Traffic Sign Recognition Benchmark (GTSRB) [3], Mapillary Traffic Sign Dataset (MTSD) [1], Tsinghua-Tencent 100K (TT100K) [4], provide comprehensive coverage for their regions. However, they are limited for specific countries and do not effectively represent the traffic sign used in Cambodia [1], [3], [4].

In Cambodia, the availability of publicly accessed Khmer traffic sign datasets remains limited [5]. Existing research on Cambodian traffic signs focuses only on seven classes [5], which do not fully represent the wide range of usually found in Cambodia roadways. Therefore, this limitation highlights a significant research gap in the development of a complete KTSDC system.

To address this research gap, this study proposes a KTSDC system based on a custom traffic sign dataset that was collected and annotated from various locations across Cambodia. The proposed collection consists of 42 classes of traffic sign types collected under various real-world conditions, including different lighting, different angles, distances, and backgrounds. Furthermore, YOLOv8 and YOLOv10 were trained and evaluated to find out the effectiveness for Khmer traffic sign detection and classification.

2 Dataset

In this section, we will describe the custom Khmer traffic sign dataset produced in this project. The dataset was manually collected and annotated for traffic sign detection and classifi-

cation in Cambodia road environments.

2.1 Dataset Overview

This paper introduces custom Khmer traffic sign dataset designed for traffic sign detection and classification in Cambodia road conditions. The proposed dataset was collected and annotated by our team, which consisted of 42 classes that consisting several categories including regulation, warning, required, and informational signals. The collected images demonstrate various real-world environments with differences in lighting, angles, distances and backgrounds. Figure 1 demonstrates the complete set of traffic sign classes included in the proposed dataset. In the next section will detail the data collection, annotation and data statistics.



Figure 1. The 42 traffic sign classes included in the proposed dataset

2.2 Data Collection

The traffic sign images were manually collected by our team from several sites across Cambodia using a smartphone camera. Data collecting was conducted on a few different road types, which include urban streets and rural streets. Images were taken under different real-world conditions, such as different lighting conditions, viewing angles, object distances, and background environments, to boost dataset diversity. After data gathering, all the images were carefully reviewed to remove duplicate images, images without traffic signs, and images that contain poor visibility that prevented clear identification of the traffic signs. This process ensured that only the quality images were left for data annotation process.

2.3 Data Annotation

Following data collection, all the images were manually annotated using the CVAT annotation tool. Each traffic sign was allocated a rectangular bounding box and provided with a class label

according to one of the 42 predefined traffic sign classes. The annotations were exported in the YOLO format, containing the class identifier and normalized bounding box coordinates for each signs. To ensure annotations quality, all annotations have been manually validated by our team, and incorrect class labels or inaccurate bounding boxes were fixed before finish datasets.

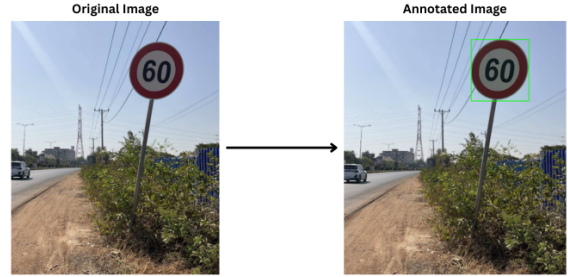


Figure 2. Example of the traffic sign annotation process. Original image. Annotated image with bounding box and class label generated using CVAT

2.4 Data Statistics

The proposed dataset consisted of 31667 images illustrating 42 classes and 31734 for annotated traffic signs, while 123 images are the images contain backgrounds images without traffic signs annotation. The number of annotated images exceeds the number of images because some images contain multiple annotated traffic signs. The distribution of annotated across the 42 classes demonstrated in Figure 3, displaying the variance in each class represent within the dataset. To simplify model training, validation and testing. The dataset was divided into training, validation, and testing subsets using a [70:20:10] ratio. The trainging subset was used to enhance the model parameters, while validation was utilized to check model performance throughout training. The testing subset designed for the final evaluation of the train models. The number of propotion of images assigned to each subset are summariezed in Table 1.

3 Methodology

The following section discusses the techniques utilized to develop and create the Khmer traffic sign detection and classification approach. The overall research process and experimental procedures are describe in the following section.

Table 1. Distribution of Images and Annotations in the Proposed Dataset

Dataset Subset	Total	Training	Validation	Testing
Images	31667	22148	6318	3201
Labels	31544	22064	6292	3188
Bounding Box	31734	22195	6335	3204

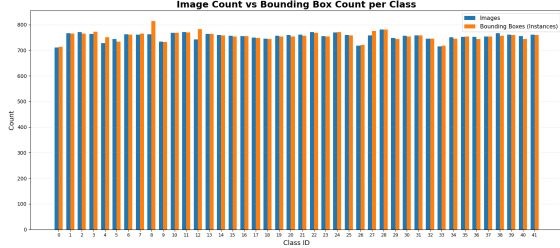


Figure 3. Class distribution of images and annotated traffic signs across the 42 classes.

3.1 Methodology Workflow

The proposed methodology establishes a systematic workflow for Khmer traffic sign detection and classification. The process begins with data preparation and data preprocessing for the annotated traffic sign dataset, followed by the model development and training under experimental set up. After training, the models will be evaluated on the test dataset using standard object detection metrics. The obtained results are subsequently compared to identify the model that achieves performance for the Khmer dataset.

3.2 Data Preprocessing

This stage was conducted to ensure that the dataset was properly prepared and the images and annotations were accurate and suitable for model processing.

3.2.1 Data Cleaning

The collected dataset was first reviewed to delete unnecessary or low quality images. Duplicates were detected and eliminated to prevent repeated samples from introducing bias into the training dataset. Furthermore, the images with severe blur were identified and removed when the traffic sign was not clearly recognizable.

The annotation files were also checked for duplicate or inconsistent labels. The image and label were reviewed to ensure that each image was associated with the correct annotation in-

formations. These cleaning methods were conducted before the annotation verification and model training stages.

3.2.2 Annotation and Label Verification

Following data cleaning, the annotation and matching class labels were verified to ensure their validation and consistency. The bounding boxes were checked to confirm that they accurately involved the traffic sign without unnecessary background regions. The images containing several traffic signs were annotated with a separate bounding box for each sign. Images without traffic signs that were identified as background were kept. The class labels were checked against the class ID in each annotation to ensure both images and labels are consistent.

3.3 Model Development

Two YOLO-based object detection models, YOLOv8 and YOLOv10, were trained on the proposed Khmer traffic sign dataset to detect and classify 42 traffic sign classes. The development process involved configuring each model for the proposed detection task, training the models using the prepared dataset, and evaluating the model performance under consistent experimental conditions. The architecture of the two models is described in the following subsections.

3.3.1 YOLOv8 Architecture

YOLOv8 is a one-stage, anchor-free object detection model applied for Khmer traffic sign detection and classification, with its overall architecture pipeline represented in Figure 4. The input images are resized to 832 x 832 before processing. The backbone uses convolutional layers, C2f modules, and Spatial Pyramid Pooling-Fast (SPPF), which extract features at different spatial resolutions. The extracted feature maps are passed to the neck, which uses an FPN-PAN structure to perform multi-scale feature fusion through upsampling, concatenation, and convolutional operations. Finally, the anchor-free, decoupled detection head generates predictions at

three feature levels, P3, P4, and P5, with strides of 8, 16, and 32, respectively, for bounding-box regression and classification [6], [7].

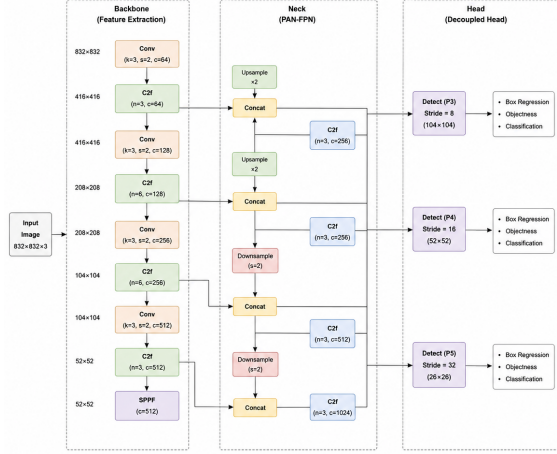


Figure 4. YOLOv8-based architecture for Khmer traffic sign detection and classification

3.3.2 YOLOv10 Architecture

The overall architecture of YOLOv10 designed for Khmer traffic sign recognition is presented in Figure 5. YOLOv10 is designed as an end-to-end object detector that avoids the Non-Maximum Suppression (NMS). In this study, the images are downsized to 832 x 832 before process by the network. The backbone utilizes efficient design including Compact Inverted Block (CIB) modules and Spatial-Channel Decoupled Downsampling (SCDown) to enhance the computing performance while maintain an important feature information. In advanced stages, the model integrates Partial Self-Attention (PSA) to enhance global feature representation with minimum computational cost. The neck performs multi-scale feature fusion using a PAN-based structure to merge semantic and Spatial Information from differnt features levels. Ultimately, YOLOv10 adopts a consistent dual-assignment method with one-to-many and one-to-one prediction head during training. One-to-many head provides the model more training information for each object, while the one-to-one head is preserved during inference to provide the final prediction without require NMS [8].

3.4 Model Training

The experiment were conducted using Google Colab with an NVIDIA Tesla T4 GPU (15GB

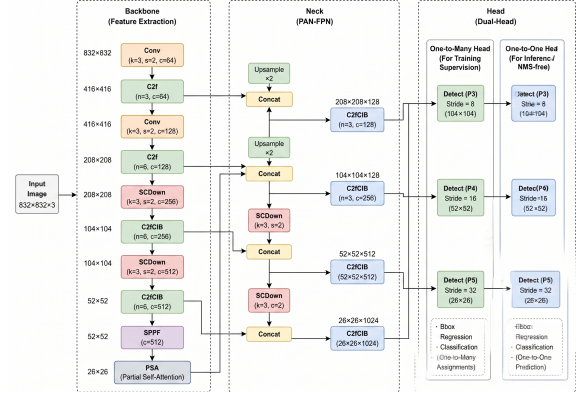


Figure 5. YOLOv10-based architecture for Khmer traffic sign detection and classification

VRAM) to train the YOLOv8 and YOLOv10 models. Both models were develop using the Ultralytics framework, established with their pre-trained weights, and fine-tuned on the prepared Khmer traffic sign dataset. The dataset for this experiment was split into 70% for training, 20% for validation and 10% for testing as shown in the Table 1.

Both model were trained with an input image size of 832 x 832 pixels, and a batch size of 6 for a maximum of 40 epochs. The AdamW optimizer was used for model optimization, with validation performance during training to monitor the model performance. The detailed training configuration and hyperparameter are summarized in Table 2.

Table 2. Training Configuration and Hyperparameters

Hyperparameter	YOLOv8	YOLOv10
Image Size	832 × 832	832 × 832
Batch Size	6	6
Epochs	40	40
Optimizer	AdamW	AdamW
Learning Rate	0.001	0.001
Weight Decay	0.0005	0.0005
Workers	2	2
Patience	20	20

3.5 Model Evaluation

Model Evaluation is the process of using the different evaluating method to analyze machine learning model performance. As shown it is a crucial step for machine learning process, as it determines the models ability to generalize

to unseen data and make accurate predictions. In this research, the models were evaluated on unseen data standard object detection metric to evaluate their ability for recognize Khmer traffic sign.

The evaluation metric used in this study include:

Accuracy: is the propotion of correct predictions among the total number of predictions made by the model.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN} \quad (1)$$

Precision: is the propotion of correctly detected among all the traffic signs predicted by the model.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

Recall: is the ratio of correctly detected object among all actual objects in the dataset.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

F1-Score: is an evaluation metric that combines precision and recall to measure the overall performance of the model.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

Intersection over Union (IoU): is measures the overlap between the predicted bounding box and the ground-truth bounding box.

$$\text{IoU} = \frac{\text{Area of Intersection}}{\text{Area of Union}} \quad (5)$$

mean Average Precision (mAP): is an evaluation metric used to measure the overall performance of an object detection model, often given as mAP@50 and mAP@95, by calculating the average precision (AP) overall traffic sign classes at different thresholds.

$$\text{mAP} = \frac{1}{N} \sum_{i=1}^N \text{AP}_i \quad (6)$$

In this research, True Positive (TP) represent a correctly detected traffic sign, True Negative (TN) represents a correctly identified non-traffic-sign region, False Positive (FP) represents an incorrect detection and False Negative (FN) represents a traffic sign that is missed by the model [9].

4 Result and Discussion

This section will discuss the experimental results of YOLOv8 and YOLOv10 on the proposed khmer traffic sign dataset. The results are compared across 20, 30 and 40 training epoch to analyze the effect of training for both models.

4.1 Overall Model Performance

Table 3 presents the performance of YOLOv8 and YOLOv10 at 20, 30, and 40 training epochs using Precision, Recall, F1-Score, mAP@50, and mAP@50-95. Both models achieved high detection performance. YOLOv8 obtained the highest Precision (96.58%), F1-Score (95.28%), and mAP@50 (98.18%), while YOLOv10 achieved the highest mAP@50-95 (77.79%). Overall, both models performed similarly well on the Khmer traffic sign dataset.

The outcomes of different training epochs on the performance of YOLOv8 and YOLOv10 is demonstrate in Figure 6. For YOLOv8, mAP@50 increased from 97.58% at 20 epochs to 97.99% at 30 epochs and reached 98.18% at 40 epochs. Furthermore, the mAP@50-95 improved from 76.70% to 77.32% and reached 77.53%. YOLOv10 also showed a consistent performance, with mAP@50 increasing from 97.14% at 20 epochs to 97.64% at 30 epochs and 97.98% at 40 epochs. Additionally, mAP@50-95 also increased from 76.83% to 77.39% and 77.79%, as represent in Table 3.

These results indicate that increasing the training epochs from 20 to 40 improved both models, while performance increase gradually become smaller, indicating stable performance.

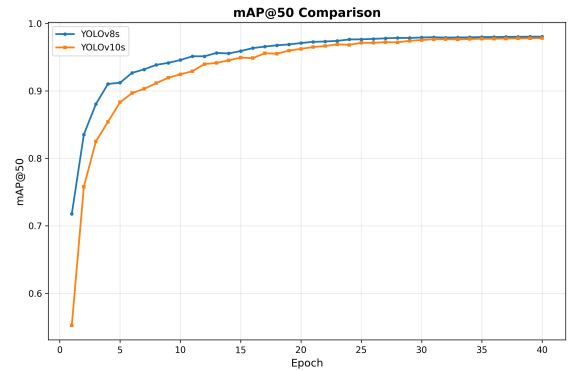


Figure 6. mAP@50 learning curves for YOLOv8s and YOLOv10s across 40 epochs.

Table 3. Performance comparison of YOLOv8 and YOLOv10 at different training epochs.

Model	Epoch	Precision (%)	Recall (%)	F1-Score (%)	mAP@50 (%)	mAP@50-95 (%)
YOLOv8	20	94.54	95.81	95.17	97.58	76.70
YOLOv8	30	95.92	95.27	95.59	97.99	77.32
YOLOv8	40	96.58	95.28	95.93	98.18	77.53
YOLOv10	20	93.86	94.16	94.01	97.14	76.83
YOLOv10	30	95.61	94.69	95.14	97.64	77.39
YOLOv10	40	95.64	95.18	95.41	97.98	77.79

4.2 Performance Evaluation of YOLOv8 and YOLOv10

As shown in Table 3, YOLOv8 demonstrated slightly stronger overall performance than YOLOv10 on the Khmer traffic sign dataset. YOLOv8 achieved stronger results across most of the evaluated metrics, demonstrating its effectiveness in detection traffic signs. However, YOLOv10 showed comparable performance and demonstrated a little advantage in mAP@50-95, indicate that YOLOv10 perform better under more stringent IoU thresholds. Overall, although the performance difference between the two models was slightly small, YOLOv8 proved a little advantage and can therefore be considered the better performance model for proposed dataset.

4.3 Detection Performance Analysis

Figure 7 demonstrates the detection and classification results generated by the selected YOLOv8 model under different conditions. The model successfully detected and classified traffic signs, these results also show that the model can recognize small or distant traffic signs and provide reliable predictions under challenging conditions, including bright lighting, low lighting and blurred images. These results provide further evidence of the model’s detection capability

5 Conclusion

This study explore the application of YOLO-based object detection for detected and classified Khmer traffic sign. The experimental results on the Khmer dataset demonstrated both YOLOv8 and YOLOv10 models achieved strong detection performance, with YOLOv8 showing a small overall advantage across the evaluation measure. YOLOv8 was able to detect and classify Khmer traffic signs even when the signs had



Figure 7. Qualitative detection results of YOLOv8 under different conditions: (a) single traffic sign, (b) multiple traffic signs, (c) small or distant traffic sign, and (d) challenging detection condition.

different sizes and the images had different lighting conditions and image quality.

This study contributes to the development of Khmer traffic sign recognition by evaluating modern object detection models using a Khmer traffic sign dataset. The results demonstrate the potential of deep learning for automatically detecting and classifying Khmer traffic signs. Furthermore, this study provides a useful reference for future research and the development of computer vision-based traffic sign recognition systems for the Cambodian road environment.

For future work, the Khmer traffic sign dataset can be expanded by including more traffic signs images within each classes and incorporating

additional traffic sign classes to create more comprehensive dataset. Future research could also explore other advanced and lightweight object detection models to enhance detection performance in real-world conditions including weather conditions, lighting environments, and road locations.

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